



POLITECNICO
MILANO 1863

EMOTIONAL OBJECTS

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PROGETTO DI INGEGNERIA INFORMATICA

Course Code: 089020

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Abstract

The “Emotional Objects” project explores the intersection between pragmatic product design and robotic art. The goal of this project is to take an everyday object and turn it into an “emotional” entity, capable of simulating vitality through expressive actions and situational reactions.

AIRLAB
ARTIFICIAL INTELLIGENCE AND ROBOTICS LAB

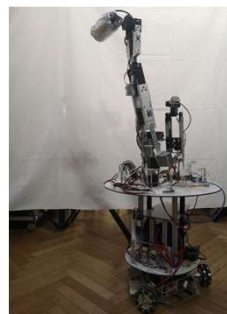
The research and fabrication were carried out within the Artificial Intelligence and Robotics Laboratory (AIRLab) at Politecnico di Milano, under the academic supervision of Professor Andrea Bonarini (Full Professor at the Department of Electronics, Information, and Bioengineering - DEIB).

Related Work



«The creation and development of an emotional robot inspired by Japanese art: R-Oiwa»,
Giovanni Cappellari

«Emotion-Driven Architecture for Human-Robot Interaction»,
Eleonora Righi



«Odile: an expressive robotic agent for emotional exchange and information sharing», Erica Panelli



1. Object Selection

Tabletop grandfather clock

2. Emotional Design

Poltergeist-like entity with multiple emotional states

3. Feasibility Analysis

Removal or rework of initial features

4. Structural and Functional Design

Details in next slide

5. Requirements Analysis

Definition of a feature check list for future reference

6. Laboratory Access

Completion of mandatory safety courses

7. Product Prototyping

Fabrication of the cardboard casing

8. Hardware Prototyping

Testing of segmented architectural units

9. Software Development

Implementation of concurrent architecture

10. Integration

Structural and software fixes in assembled product

11. Finalization

Aesthetic improvement and supervisor demonstration



Structural and Functional Design

The design is divided into two sections:

- **Head**

- The charging holes house LEDs to represent glowing eyes.
- The hands can either spin counterclockwise or stop at set positions to simulate expressions, like furrowed eyebrows or a mustache.
- The brand, usually written on the dial, is placed on a rotating disk as a curved inscription, which can appear either as a smile or a pout.
- The lunar dial can rotate to show different images associated with various emotions: a sunny sky for happiness, stormy clouds for anger, and a starry night for calm or sadness.

- **Body**

- A base equipped with an opening compartment, which the clock will autonomously open to encourage interaction during specific emotional states. It contains colored buttons that unlock different reactions based on the current emotion.
- A highly mobile pendulum. In addition to normal oscillations at various speeds, it can perform a curving motion as if reaching towards the user.



Hardware

Microcontroller Selection

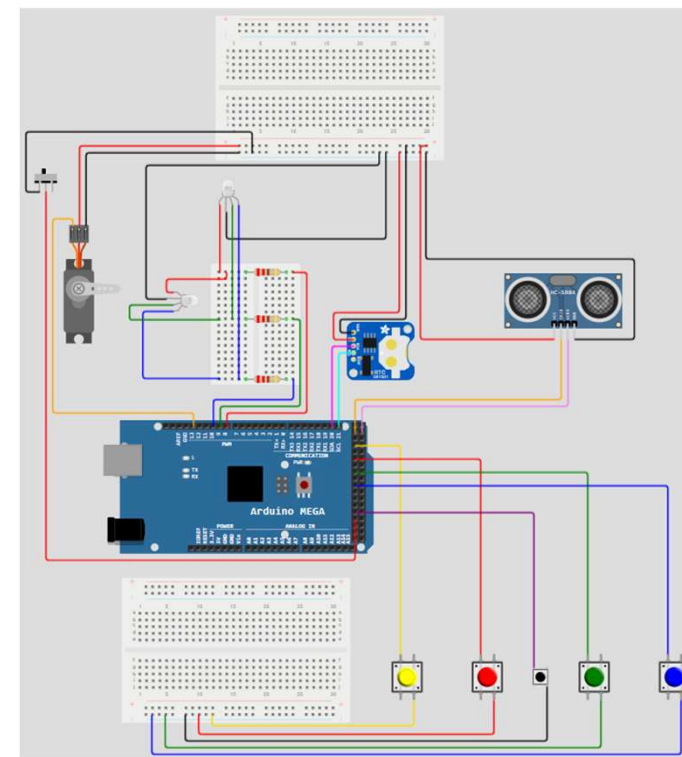
The Arduino Mega 2560 was selected as the optimal solution for the project thanks to its elevated PWM pin count, high-performance processing capabilities, and native 5V operating voltage.

Sensors

- **HC-SR04 Ultrasonic Sensor:** Proximity detection.
- **Digital Pushbuttons:** User input.
- **Microswitches:** Mechanical triggers for state detection and homing logic.
- **A3144 Hall Effect Sensor:** Non-contact trigger for homing logic.
- **DS3231 RTC Module:** Accurate timekeeping with external battery.

Actuators

- **SG90 Positional Servos:** Precise angular control for dial disks, pendulum, and compartment latch.
- **FS90R Continuous Rotation Servos:** Seamless clock hand movement.
- **Common-Cathode RGB LEDs:** Color variation for the eyes with full color spectrum.



Simplified Circuit Diagram



Software

The software was coded within the Arduino IDE, a developing environment specialized in compiling and uploading code to Arduino hardware, equipped with real-time serial communication tools needed for debugging.

The program utilizes C programming language low-level procedural paradigms, mixed with object-oriented C++ syntax to communicate with the following peripheral drivers:

- **Servo.h:** Handles PWM signal generation for precise degree-based motor control.
- **NewPing.h:** Offers non-blocking ultrasonic echo timing to ensure concurrent system movements.
- **Wire.h & RCTlib.h:** Manages background I²C protocol communication for accurate and formatted timestamps.

Workshop Equipment

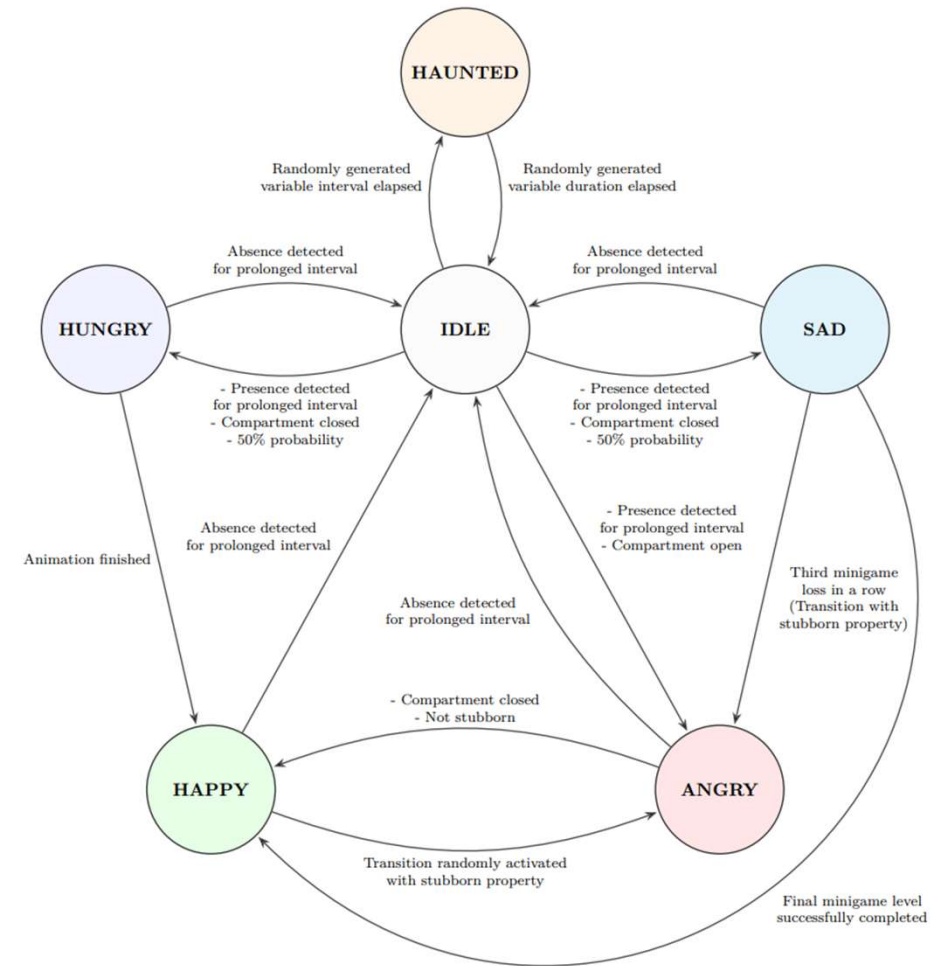
Beyond standard crafting utensils (e.g., box cutters, pliers, screwdrivers, hand saws, files, etc.), the assembly process required specialized equipment found in the AIRLab workshop, specifically a drill press and a soldering station, whose use warranted detailed explanation.

The complete source code, accompanied by an operational manual, is hosted in the project's official GitHub repository, which can be accessed at the following URL:

<https://github.com/michelcostantini/Emotional-Objects-Tempus-Mordet>

The global architecture is structured like a layered finite state machine. The concurrent nature of the updates allows the definition of specialized state machines.

- **Core FSM:** Coordinates general system transitions based on randomized events or user detection.
- **Emotional Sub-States:** Execute the emotional behavior described in the state diagram to the right.
- **Nested Memory Game FSM:** During the sad emotion, handles sequence generation and validation logic, user input, and visual animations.
- **Mechanical Hand Controller:** A hardware-level FSM designed to govern hand movement and positioning during ticking, homing, offsetting, and waiting states.





Non-Blocking Design

Since the object behavior required concurrent movement without losing access to sensor data, the software development had to move away from blocking delay() calls. Thus, a non-blocking paradigm that relies on the millis() function was adopted.

The program implements an asynchronous scheduler which cycles through tasks and confronts their last update timestamp with the current time. If the fixed execution interval has elapsed, the task gets updated.

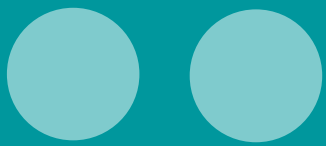
This allows smooth execution and optimizes power consumption.

Coaxial Movement

To simulate the overlapping layout of clock hands while preserving independent movement, the mechanism had to adopt a custom pulley-driven transmission system.

Since the motors cannot be stacked on top of one another, only one servo was kept directly aligned with the central rotation axis, while the other was dislocated diagonally. The rotational torque of the dislocated servo was transferred to the central axis using two identically sized pulleys (1:1 ratio) linked by an elastic rubber band.

While this successfully achieved coaxial movement, it also introduced complexities in the hand homing logic.



Testing

The system's fulfillment of the requirements was verified by observing a full execution sequence upon booting the microcontroller. The prototype followed the intended behavior:

- Initialization with error management.
- Asynchronous actuator and sensor tasks using non-blocking scheduling.
- Continuous proximity and absence detection.
- Emotional behavior with conditional and randomized state transitions.
- Expressiveness provided by multiple visual clues.
- Mini-game execution specific to the sad state.
- Happy state user controls handling.

User Experience

After completion, the product was presented to new users whose interactions served as additional testing and provided useful perspectives on design choices.

The object proved to effectively ignite user curiosity and attention.

The emotional states and their expressive representation were positively received by the users.

The users experienced varying interaction sequences, witnessing the full behavior after multiple approaches.

Although intentionally hard to decipher, the needs of the clock were ultimately understood by the users who managed to satisfy them, unlocking new reactions.



Emotional Expressions



Happy State



Angry State



Sad State



*Hungry State detail:
curved pendulum*